CHL7201/CLL788: Assignment 2 - RNN, Encoder-Decoder, Attention and VAE

Report for Problem 5

This report describes the coding experiments. All networks were implemented in PyTorch. Random seed = 0.

The notebook sets both torch.manual_seed(0) and np.random.seed(0) before training, and every table below is taken from that run. The code is CHL7201_Assignment2.ipynb, section Q5. Loss during training is mean squared error on the scaled target. Every RMSE below is converted back to the original units, percent silica.

1. Data preparation

The file is the flotation-plant database. It has 737,453 rows and 24 columns, from 10 March 2017, 01:00, to 9 September 2017, 23:00. The numbers use a decimal comma, so the file is read with decimal=','. A typical hour contains 180 rows, one sample every 20 seconds. Averaging inside each clock hour leaves 4,097 hours.

% Iron Concentrate is a second laboratory assay of the same product as % Silica Concentrate. It is dropped before any model sees the table, so the network cannot read the answer off a paired lab measurement. That leaves 21 process columns. Silica is the target. In the soft sensor, silica is not an input. In the forecaster, past silica is an input, because the task is to continue the silica series.

There is one break in the clock. 16 March 2017, 05:00 jumps to 29 March 2017, 12:00, a gap of 13 days and 7 hours. Hours are grouped into contiguous segments, and any window that would step across that shutdown is discarded. The same rule keeps a window from mixing training, validation and test hours.

The hours are cut in time order, 70/15/15, which is 2,867 / 614 / 616 hours. The scaler is fit on the training hours only. On those hours the silica mean is 2.3176 and the standard deviation is 1.1256. Predicting that training mean on the L = 12 test windows gives RMSE 1.2108. That is the baseline the soft sensors have to beat.

2. Soft sensor

The soft sensor predicts silica at the next hour from the previous L hours of the 21 process variables. Three models share the same shape: one recurrent layer with 32 units, then a linear output. The vanilla RNN uses tanh. Optimiser is Adam with learning rate 1e-4, for 20 epochs, batch size 128. Gradients are clipped at norm 1. The same budget is used for every model and every window length.

At L = 12 the test errors are:

Model	Parameters	Test RMSE
Vanilla RNN	1,793	1.1837
LSTM	7,073	1.2264
GRU	5,313	1.2333
Training-mean baseline	—	1.2108

The vanilla RNN is the only one of the three that beats the baseline at this window. The LSTM and the GRU have several times as many parameters and a higher test error. Extra gating is not helping this one-hour target.

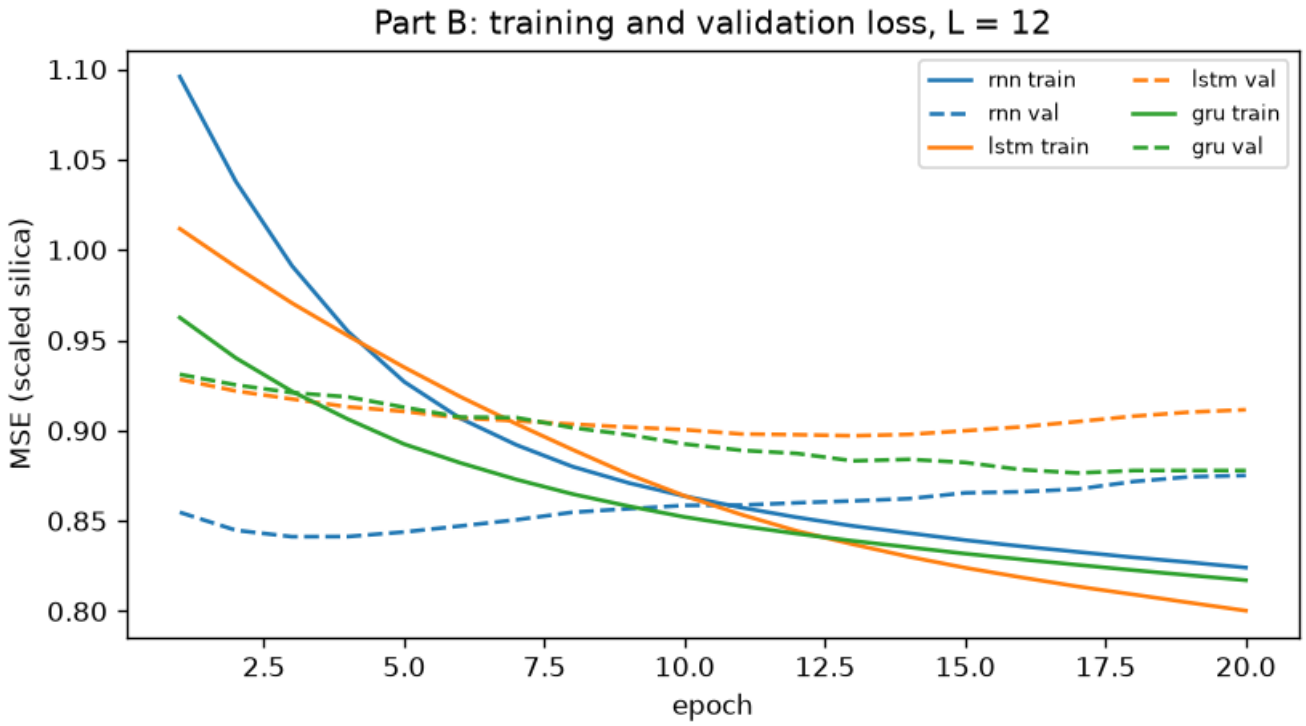


Figure 1. Training loss (solid) and validation loss (dashed) at L = 12, in scaled silica units. Training loss falls for all three models. Validation loss flattens. Twenty epochs is enough to see the comparison, and it is the same number of epochs for each model.

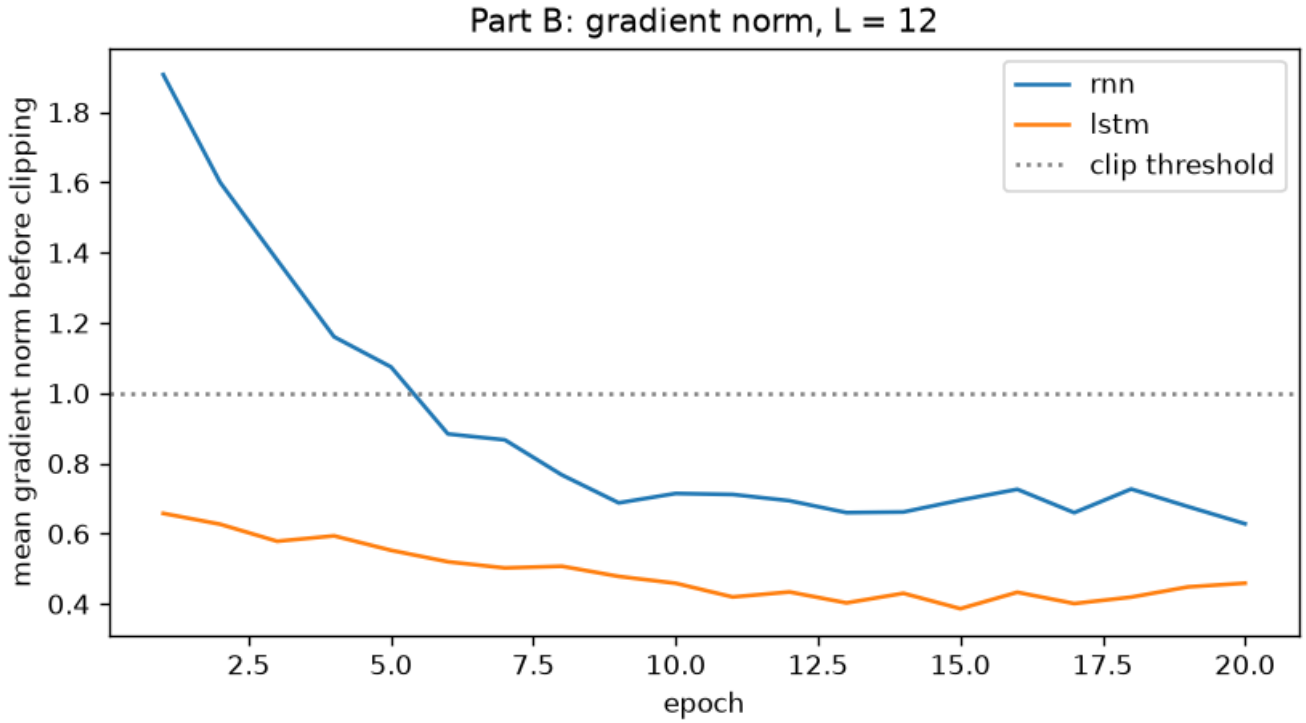


Figure 2. Mean gradient norm before the clip, at L = 12. The dotted line is the clip threshold of 1.

At epoch 1 the RNN norm is 1.907, so the clip of 1 binds and the step is cut down. By epoch 20 the RNN norm is 0.628 and the clip no longer binds. The LSTM starts at 0.658 and ends at 0.459, under the clip the whole way. Clipping can stop a gradient that has already grown too large. It cannot bring back a gradient that has vanished along a long chain. The next plot is where that shows up.

The same two models were retrained at L = 6, 24 and 48, with the same hidden size, learning

rate, epochs and clip.

L (hours)	RNN test RMSE	LSTM test RMSE
6	1.1790	1.2140
12	1.1837	1.2264
24	1.1919	1.2392
48	1.1976	1.2562

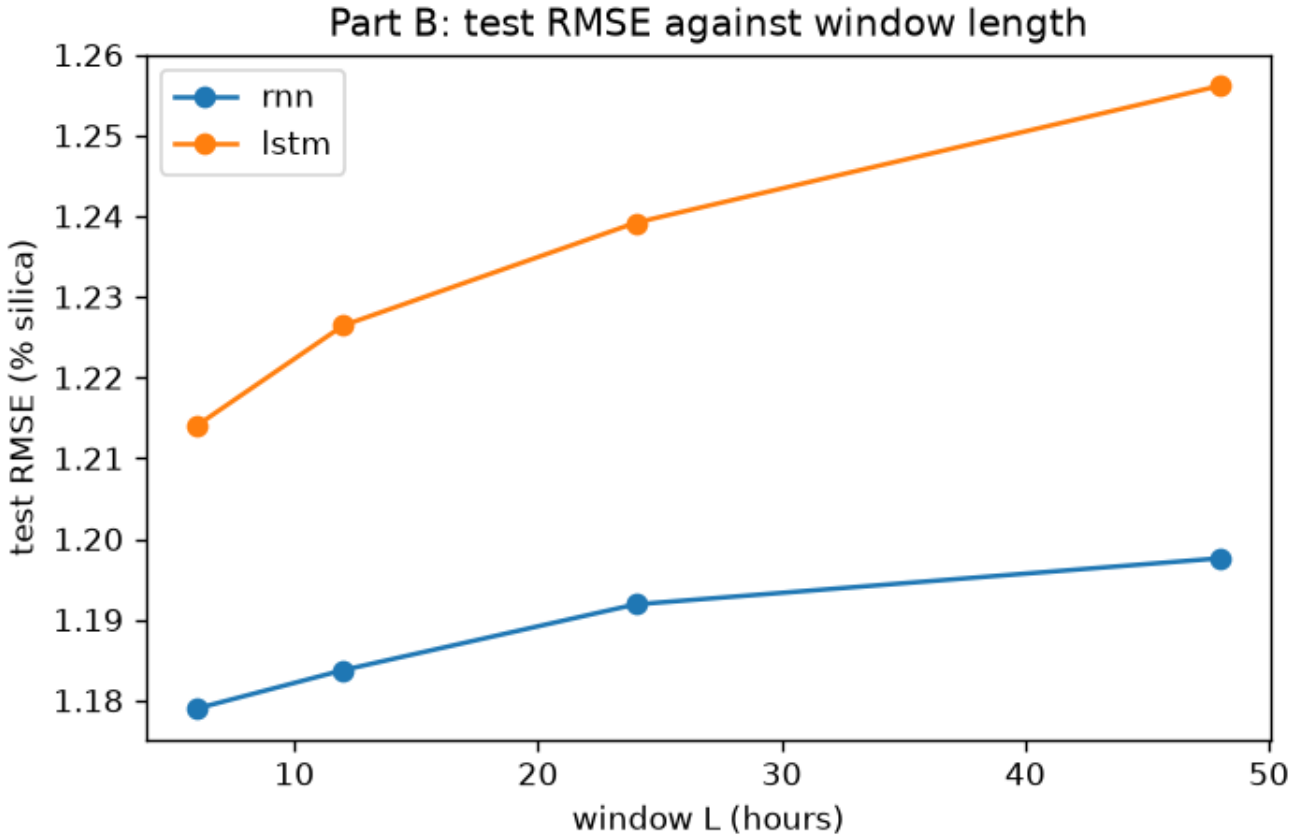


Figure 3. Test RMSE against the length of the input window.

A longer memory does not lower the error. From L = 6 to L = 48 the RNN moves from 1.1790 to 1.1976, and the LSTM moves from 1.2140 to 1.2562. Each extra RNN step multiplies the backward signal by at most |W_hh| tanh', which is smaller than 1, so an hour from two days ago arrives with almost no gradient. On this plant the useful signal for the next silica reading is recent. The soft sensor to deploy is the vanilla RNN with L = 6 (test RMSE 1.1790).

3. Six-hour forecast

The forecaster reads the previous 24 hours of the 21 process variables together with the past silica, so each input step has 22 numbers. The target is the next 6 silica values. A window that would cross the March shutdown, or that would mix the train, validation and test blocks, is dropped.

The encoder is an LSTM with 32 units. Its final state (h_T, c_T) is the context, and it is also the decoder's starting state. The decoder is an LSTM that reads one silica number and writes one prediction. Its first input is the last silica that was actually measured. Under teacher forcing, every later input is the true previous silica. Under inference, every later input is the decoder's own previous prediction.

Both forecast models use Adam with learning rate 1e-3, 10 epochs, batch size 128, and the

same clip of 1. The plain encoder-decoder has 11,681 parameters. Its training MSE goes from 0.823 to 0.371 and its validation MSE from 0.747 to 0.365, so at 10 epochs the two curves are still together.

Test RMSE at each hour ahead:

Hour	Teacher forcing	Inference	Attention	Repeat last
1	0.8641	0.8641	0.8416	0.7566
2	0.7907	0.9933	0.9511	0.9255
3	0.7640	1.1059	1.0422	1.0511
4	0.7482	1.1889	1.1178	1.1668
5	0.7383	1.2342	1.1672	1.2473
6	0.7324	1.2541	1.1970	1.3055

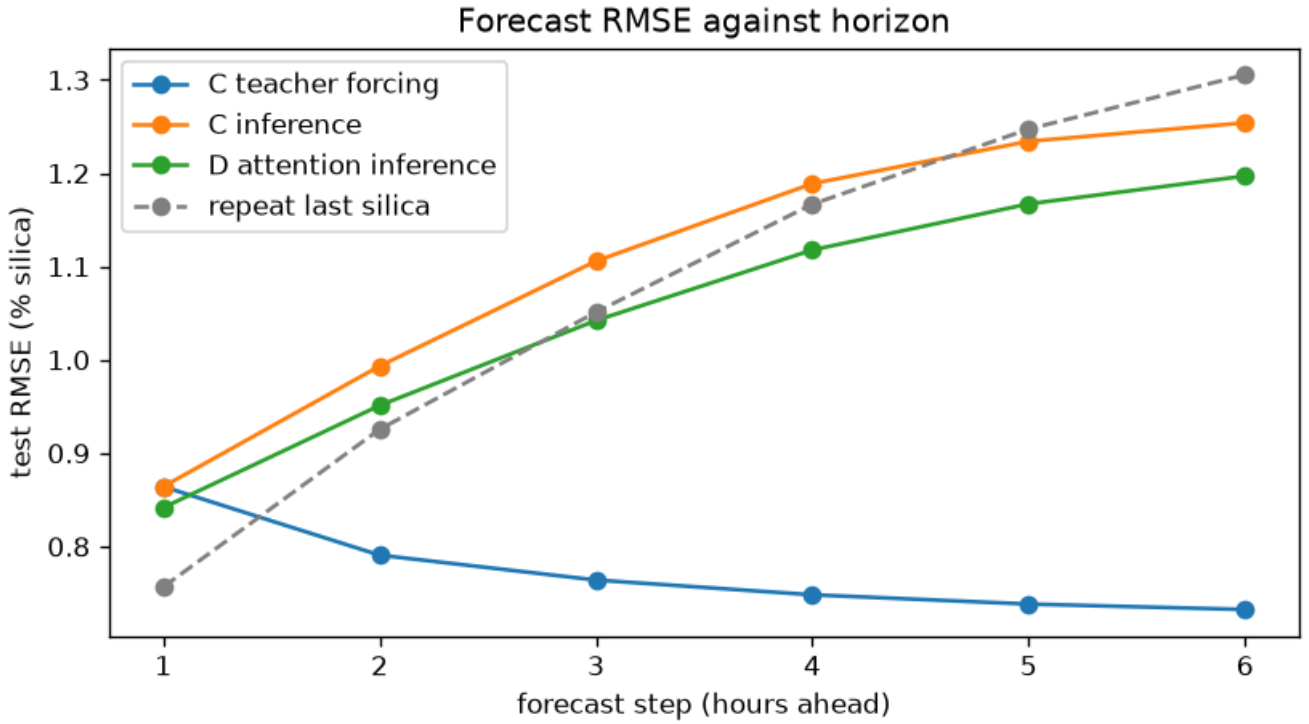


Figure 4. Test RMSE against the forecast horizon. Teacher forcing stays low. Inference rises. Attention inference rises more slowly, and from hour 3 it is also below repeating the last silica.

At one hour ahead, teacher forcing and inference are the same number, 0.8641. Both of them start from the measured silica, so there is no earlier prediction to feed back yet. From hour 2 they separate, and the gap is larger at hour 6 than at hour 2. Teacher forcing is allowed to see the true silica at every step, so by hour 6 the decoder has been shown five true values and the prediction is still a one-step prediction from a correct history. Inference writes the hour-1 prediction back in as the input for hour 2. If hour 1 is a bit wrong, hour 2 starts from the wrong place, and hour 3 starts from a place that is wrong for two reasons. The error piles up.

Repeating the last measured silica is stronger than either network at one hour (0.7566) and weaker than both networks by six hours (1.3055). A one-hour forecast on this plant is close to persistence. A six-hour forecast is not.

4. Attention

The attention model uses the same encoder, the same horizon, and the same training budget.

Before each decoder step the score of encoder state i is the dot product of the current decoder state with that encoder state. A softmax turns the scores into weights that add up to 1, and the context is the weighted sum of the 24 encoder states. A matrix W_c sends the 32-dimensional context down to 8 numbers, and those 8 numbers are joined to the one silica input before they enter the decoder. That keeps the laboratory value from being drowned by the context vector. The model has 12,961 parameters. Training MSE goes from 0.773 to 0.371 and validation MSE from 0.738 to 0.338.

Under inference, attention is below the plain encoder-decoder at every horizon: 0.8416, 0.9511, 1.0422, 1.1178, 1.1672, 1.1970. From hour 3 it is also below repeating the last silica. The six-hour forecaster to deploy is this attention model. Its inference RMSE at 6 hours is 1.1970, against 1.2541 with no attention and 1.3055 for persistence.

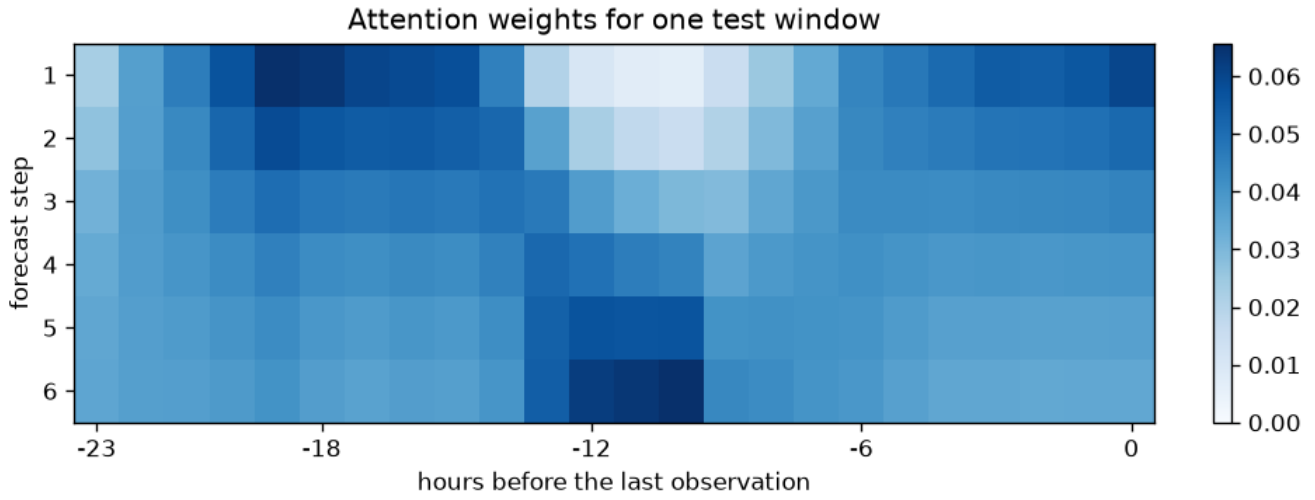


Figure 5. Attention weights on one test window, under inference. Darker means a larger weight. The horizontal axis is hours before the last observation. An equal share of the 24 hours would be 1/24 = 0.0417.

On this window the largest weight is 19 hours before the last observation for forecast steps 1, 2 and 3, then 13, 12 and 10 hours before it for steps 4, 5 and 6. The weights are spread out rather than piled on one hour. The mean weight on the most recent hour is 0.0445, and the mean weight on the oldest hour is 0.031. The window leans on a band about 10 to 19 hours back, and that band shifts toward the present as the forecast steps forward. It does not put all of its weight on the single most recent hour.

5. What to deploy

The discussion below is 150 words. It is the paragraph written by the notebook from the tables in this report.

Deploy the vanilla RNN soft sensor at L = 6 (test RMSE 1.18; training-mean baseline 1.21). The L = 12 LSTM (1.23) and GRU (1.23) are higher. From L = 6 to L = 48 the RNN error rises to 1.20, since each backward step is scaled by at most |W_hh| tanh' and distant hours fade. The RNN gradient norm starts at 1.9, so the clip of 1 binds; the LSTM starts at 0.7. Deploy the attention encoder-decoder as the 6-hour forecaster. Inference RMSE at 6 hours is 1.20, versus 1.25 with no attention and 1.31 for repeating the last silica. Teacher forcing stays between 0.73 and 0.86 while inference rises from 0.86 to 1.25, as each prediction is fed back. Plotted weights peak 10-19 hours back. A VAE with epsilon = 0 on the 21 process variables should flag hours whose reconstruction error is many times the normal value.